

SELECT with various clause – BETWEEN, IN, Aggregate function

IN & BETWEEN:

1)

```
mysql> select course_type
-> from course
-> where course_name in ('Statistics','Programming');
+-----+
| course_type |
+-----+
| Core        |
| Core        |
+-----+
2 rows in set (0.00 sec)
```

2)

```
mysql> select course_no
-> from registration
-> where score between 50 and 80;
+-----+
| course_no |
+-----+
| C001      |
| C001      |
| C002      |
| C002      |
| C002      |
| C002      |
+-----+
6 rows in set (0.00 sec)
```

AGGREGATE FUNCTIONS:

1)

```
mysql> select avg(score) as average_mark
-> from registration
-> where course_no='C002';
+-----+
| average_mark |
+-----+
|      68.4000 |
+-----+
1 row in set (0.00 sec)

mysql>
```

2)

```
mysql> select max(score) as maximum_mark,
-> min(score) as minimum_mark
-> from registration
-> where course_no='C002';
+-----+-----+
| maximum_mark | minimum_mark |
+-----+-----+
|           82 |           55 |
+-----+-----+
1 row in set (0.00 sec)
```

3)

```
mysql> select course_no,  
->      max(score) as maximum_mark,  
->      min(score) as minimum_mark,  
->      avg(score) as average_mark  
-> from registration  
-> group by course_no;
```

course_no	maximum_mark	minimum_mark	average_mark
C001	78	65	71.5000
C002	82	55	68.4000
C003	90	90	90.0000
C004	88	88	88.0000
C005	95	95	95.0000

5 rows in set (0.00 sec)

4)

```
mysql> select c.course_name,  
->      avg(r.score) as average_mark  
-> from course c  
-> join registration r  
-> on c.course_no=r.course_no  
-> group by c.course_name;
```

course_name	average_mark
Programming	71.5000
Statistics	68.4000
Database	90.0000
Mathematics	88.0000
Physics	95.0000

5 rows in set (0.00 sec)

5)

```
mysql> select sum(score) as total_score
-> from registration;
+-----+
| total_score |
+-----+
|          758 |
+-----+
1 row in set (0.00 sec)

mysql> █
```

6)

```
mysql> select c.course_name,
->          count(r.reg_no) as total_students
-> from course c
-> join registration r
-> on c.course_no=r.course_no
-> group by c.course_name;
+-----+-----+
| course_name | total_students |
+-----+-----+
| Programming |                2 |
| Statistics  |                5 |
| Database    |                1 |
| Mathematics |                1 |
| Physics     |                1 |
+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

7)

```
mysql> select s.name,  
->         count(r.course_no) as total_courses  
-> from student s  
-> join registration r  
-> on s.reg_no=r.reg_no  
-> group by s.name;
```

name	total_courses
mohan	2
raju	2
ramesh	2
ram	2
san	2

5 rows in set (0.00 sec)

```
mysql> █
```